Emmanuel Adutwum
Soka University of America
Aliso Viejo, California, United States
eadutwum@soka.edu

3 September 2026

Dr Santosh Reddy Addula, Editor
Mr Ramakant Mohite, Assistant Editor
Discover Artificial Intelligence
Springer Nature

Re: Submission ID 6cf4e30e-1a5b-4ad2-99e5-2320f0737507 — revised manuscript

Dear Dr Addula and Mr Mohite,

Please find enclosed the revised version of my manuscript, now entitled "A Controlled Reassessment of Support Vector Regression on the California Housing Benchmark with Feature Scaling, Feature Selection and Hyperparameter Tuning", submitted in response to the decision of major revision. A point-by-point response to every comment from the Editor and all three reviewers accompanies this letter, together with an updated Contribution Information Sheet.

I want to be direct about what changed, because the revision is not cosmetic. Two of the central claims of the previous version were wrong, and the reviewers found both.

First, the value of $R^2 \approx 0.60$ that the previous version attributed to the SVR of Preethi et al. is not their SVR result at all — it is the score of their linear, ridge and polynomial-ridge models. Their SVR-RBF result is approximately 0.15. Reviewer 1 and Reviewer 3 independently observed that an unscaled baseline of -0.054 could not explain a reported 0.60, and they were right: the premise was mistaken. Reproducing the configuration they describe, over 5 seeds, now yields $R^2 = -0.020$ with library defaults and 0.514 when tuned — values that bracket their actual result — while adding a single standardisation step to the identical pipeline yields 0.766. The causal story the paper tells is now supported by a direct reproduction rather than by an assumed correspondence.

Second, the previous version's claim that feature scaling accounted for 95.7% of the improvement was an artefact of evaluating a single ablation ordering. Evaluating all 32 cells of the four-factor design shows that the apparent contribution of scaling varies by 0.7441 in R^2 depending only on where it is placed in the sequence. The order-independent Shapley value is 0.2723. Scaling remains the largest single contributor, but the previous percentage is withdrawn.

The revision also reports three results that contradict the earlier framing, and that I have chosen to report rather than omit. The feature-specific scaling strategy I previously presented as the paper's preprocessing advance does *not* outperform plain standardisation on this dataset — it is worse, by a difference that is significant under a paired test. The distribution-aware scaler rule I introduced in this revision to replace it does not transfer either: applying the workflow to King County and Ames housing data, which the Editor and Reviewer 2 asked for, shows that the component ordering holds on all 3 datasets under plain standardisation but on only 1 under the rule, whose contribution on Ames is negative. Testing the workflow on further datasets therefore falsified part of it rather than corroborating it, and I have withdrawn the rule as a contribution. And the derived features contribute little once selection and tuning are in place. In addition, spatially blocked cross-validation, which Reviewer 1 requested, reveals that the optimism of random partitioning on these geographically structured data exceeds every pipeline effect measured anywhere in the study. That finding qualifies the headline numbers of this literature, including my own, and it is now stated as such.

Beyond these, the revision adds a direct reproduction of the prior configuration, an equal-budget comparison in which all 10 models receive identical data and identical tuning, corrected confidence intervals with paired significance testing on identical splits under the Nadeau–Bengio correction, feature selection repeated inside every fold together with fully nested estimates, replication of the workflow on King County and Ames housing data, and sensitivity analyses for every hand-chosen constant. The workflow itself is now defined explicitly as a staged procedure with a diagram, the methods sections have been expanded substantially, the claim of novelty has been withdrawn and replaced with a methodological and diagnostic contribution, and the manuscript has been restructured and copy-edited throughout.

On the reporting inconsistencies noted by Reviewer 1, I took a structural approach rather than a proofreading one. Every experiment now writes a JSON record, and every table, figure and in-text number in the manuscript is generated from those records by script. No numerical value is typed by hand anywhere in the paper, so the text, the tables and the figures cannot diverge.

Following the internal editorial feedback, the Declarations section now addresses ethical approval, consent to participate and consent to publish under individual subheadings; the data availability statement appears above the References with direct links to every dataset and to the code repository; and a dual publication declaration records the arXiv preprint (arXiv:2605.08660) and the relationship between it and the present manuscript. As Reviewer 2 requested and the Editor endorsed, the manuscript now includes a disclosure of my use of a large language model as an assistive tool, together with a statement of what remains my own responsibility.

I am grateful to the reviewers. The requirement to reproduce the prior configuration directly is what uncovered my misreading of the reference value, and the requirement to evaluate the full ablation design is what showed that my headline attribution depended on an arbitrary ordering. The paper is more modest than it was, and considerably more solid.

The manuscript has not been published elsewhere and is not under consideration by any other journal. I have no competing interests to declare. I would be glad to provide any further information the Editor or reviewers may require.

Thank you for your consideration.

Sincerely,

Emmanuel Adutwum
Soka University of America